



Construction and Analysis of a Nonstandard Finite Difference Scheme for Stochastic Partial Differential Equations

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Abstract:

This paper is devoted to the development and comprehensive analysis of a nonstandard finite difference (NSFD) scheme for Itô stochastic partial differential equations (SPDEs). The qualitative properties of the proposed scheme, including consistency, stability, and convergence, are thoroughly investigated. Numerical simulations are presented to demonstrate the accuracy and computational efficiency of the proposed scheme.

Keywords: Nonstandard finite difference scheme, consistency, stability, convergence, stochastic partial differential equations.

Mathematics Subject Classification (2010): 65M06, 65M12, 60H15.

1 Introduction

Over the past few decades, stochastic partial differential equations (SPDEs) have emerged as a powerful framework for modelling realistic phenomena, as they accurately reflect the fundamentally random nature of the underlying processes. SPDEs arise in a wide range of scientific disciplines, including chemical physics, fluid mechanics, biology, and economics. Consequently, this class of problems has attracted growing interest, and considerable research effort has increasingly been devoted to the analysis of SPDEs. Moreover, since the exact solutions of SPDEs are, in many cases, difficult or even impossible to obtain, the

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development of effective numerical methods for their approximation has become essential. Stochastic finite difference (SFD) schemes are among the most widely used numerical methods for approximating the solutions of SPDEs. The works [Namjoo et al. \(2024\)](#); [Baleanu et al. \(2023\)](#); [Karami et al. \(2023\)](#); [Namjoo and Mohebbian \(2016, 2019\)](#) are devoted to the convergence and stability analysis of several stochastic finite difference schemes for classes of SPDEs. . This paper develops a novel NSFD scheme for SPDEs driven by a one-dimensional white noise process. The remainder of this paper proceeds as follows. Section 2 briefly reviews the structure of NSFD schemes. Section 3 presents an explicit NSFD scheme for approximating the solution of SPDEs. In Section 4, the consistency, stability, and convergence of the proposed NSFD scheme are established. Finally, Section 5 concludes the paper and presents some numerical results.

2 Nonstandard Finite Difference Schemes for Initial Value Problems

In this section, we review the fundamental principles of NSFD schemes. Mickens introduced the NSFD framework for ordinary differential equations [Mickens \(2020\)](#); this approach has been further employed by the authors in several studies [Namjoo et al. \(2025\)](#); [Namjoo and Zibaei \(2019\)](#); [Namjoo et al. \(2018\)](#); [Namjoo and Zibaei \(2020\)](#); [Karami et al. \(2024\)](#). To illustrate an NSFD scheme, we consider the following nonlinear initial value problem

$$Y'(t) = f(Y(t)), \quad Y(t_0) = Y_0, \quad t \in [0, T_f].$$

An NSFD scheme can be constructed by employing at least one of the following two steps. We utilize the discrete approximation

$$Y'(t_k) \approx \frac{Y_{k+1} - Y_k}{\mathcal{X}(\Delta t)},$$

where Y_k denotes the approximation of $Y(t_k)$ at time step $t_k = k\Delta t$ for $k = 0, 1, \dots, N$. The nontrivial denominator function $\mathcal{X}(\Delta t)$ must be continuous and strictly increasing, satisfying the following criterion:

$$\mathcal{X}(\Delta t) = \Delta t + O(\Delta t^2), \quad 0 < \mathcal{X}(\Delta t) < 1, \quad \Delta t \rightarrow 0.$$

Some examples of denominator functions are given in [Namjoo and Zibaei \(2020\)](#):

$$\frac{e^{\lambda \Delta t} - 1}{\lambda}, \quad \tanh(\Delta t), \quad \sin(\Delta t).$$

Nonlocal approximations are applied to both nonlinear and linear terms; for example,

$$x(t)z(t)|_{t=t_k} \approx x_{k+1}z_k, \quad x(t)|_{t=t_k} \approx x_{k+1}, \quad z^2(t)|_{t=t_k} \approx z_k z_{k+1}.$$

3 An explicit NSFD scheme for SPDEs

In this section, we develop a nonstandard finite difference (NSFD) scheme for solving the SPDE of the form

$$u_t(x, t) + a\gamma u_{xx}(x, t) + b u(x, t) + c u(x, t)\dot{W}(t) = 0, \quad x \in [0, 1], \quad t \in [0, 1], \quad (3.1)$$

subject to the initial and boundary conditions

$$u(x, 0) = u_0(x), \quad u(0, t) = f(t), \quad u(1, t) = g(t),$$

where b and c are positive constants and $\dot{W}(t)$ denotes white noise, modelled as a Gaussian process with zero mean [Kloeden and Platen \(1995\)](#). The linear term of (3.1) is represented by the nonlocal approximation $u_k^n \approx \frac{1}{2}(u_{k+1}^n + u_k^n)$, where u_k^n denotes the numerical solution at the grid point $(x_k, t_n) = (k\Delta x, n\Delta t)$. The temporal and spatial derivatives, together with the function $u(x, t)$ appearing in the deterministic part, are discretized as

$$u_t(k\Delta x, n\Delta t) \approx \frac{u_k^{n+1} - u_k^n}{\Delta t}, \quad u_{xx} \approx \frac{u_{k-1}^n - 2u_k^n + u_{k+1}^n}{\Delta x^2}, \quad u_k^n \approx \frac{u_{k+1}^n + u_k^n}{2}.$$

Substituting these approximations into (3.1) yields the explicit NSFD scheme

$$u_k^{n+1} = -r u_{k-1}^n + (1 + 2r - s) u_k^n - (r + s) u_{k+1}^n - c u_k^n \Delta W_n, \quad (3.2)$$

where $r = \frac{a\Delta t}{\Delta x^2}$, $s = \frac{b\Delta t}{2}$ and $\Delta W_n = W((n+1)\Delta t) - W(n\Delta t) \sim N(0, \Delta t)$. Throughout the remainder of this manuscript, the increments of the Wiener process are assumed to be independent of the state u_k^n .

4 On the Consistency, Convergence, and Stability of the Proposed NSFD Scheme

Consider a SPDE of the form $Lu = G$ where L denotes the differential operator and G is source term. Let $L_h^n u_k^n = G_k^n$ denote the NSFD scheme. In order to analysis consistency and stability the NSFD scheme (3.2), it is very important to consider a norm. Hence, for a sequence $\{u_k^n\}_{0 \leq k \leq M}$, we define $\|u^n\| = \sqrt{\sup_{0 \leq k \leq M} |u_k^n|^2}$. In order to get consistency, stability and convergence results, we give the following definitions [Roth \(2002\)](#).

Definition 4.1. A NSFD scheme $L_h^n u_k^n = G_k^n$ is consistent with the SPDE $Lu = G$ at point (x, t) , if for any continuously differentiable function $\Phi(x, t)$ in the sense of mean square we have $\mathbb{E}\|u^{n+1} - v^{n+1}\| \rightarrow 0$, for $t = (n+1)\Delta t$ and $\Delta t, \Delta x \rightarrow 0$, where the vectors u^{n+1} and v^{n+1} are defined by $u^{n+1} = (u_0^{n+1}, u_1^{n+1}, \dots, u_M^{n+1})$ and $v^{n+1} = (v_0^{n+1}, v_1^{n+1}, \dots, v_M^{n+1})$.

Theorem 4.2. *The NSFD scheme (3.2) is consistent in the sense of mean square.*

Proof. Let $\Phi(x, t)$ be a smooth function, thus

$$\begin{aligned} L(\Phi)|_k^n &= \Phi(k\Delta x, (n+1)\Delta t) - \Phi(k\Delta x, n\Delta t) \\ &+ a \int_{n\Delta t}^{(n+1)\Delta t} \Phi_{xx}(k\Delta x, s) ds + b \int_{n\Delta t}^{(n+1)\Delta t} \Phi(k\Delta x, s) ds \\ &+ c \int_{n\Delta t}^{(n+1)\Delta t} \Phi(k\Delta x, s) dW(s), \end{aligned}$$

and

$$\begin{aligned} L_h^n \Phi &= \Phi(k\Delta x, (n+1)\Delta t) - \Phi(k\Delta x, n\Delta t) \\ &+ \frac{a\Delta t}{\Delta x^2} (\Phi((k+1)\Delta x, n\Delta t) - 2\Phi(k\Delta x, n\Delta t) + \Phi((k-1)\Delta x, n\Delta t)) \\ &+ \frac{b\Delta t}{2} (\Phi((k+1)\Delta x, n\Delta t) + \Phi(k\Delta x, n\Delta t)) + c\Phi(k\Delta x, n\Delta t)\Delta W_n. \end{aligned}$$

Now by using the square property of Itô integral, one can obtain the following inequality

$$\begin{aligned} &\mathbb{E}|L(\Phi)|_k^n - L_k^n \Phi|^2 \\ &= \mathbb{E} \left| a \int_{n\Delta t}^{(n+1)\Delta t} \left[\Phi_{xx}(k\Delta x, s) - \frac{1}{\Delta x^2} (\Phi((k+1)\Delta x, n\Delta t) \right. \right. \\ &\quad \left. \left. - 2\Phi(k\Delta x, n\Delta t) + \Phi((k-1)\Delta x, n\Delta t)) \right] ds \right. \\ &\quad \left. + b \int_{n\Delta t}^{(n+1)\Delta t} \left[\Phi(k\Delta x, s) - \frac{1}{2} (\Phi((k+1)\Delta x, n\Delta t) + \Phi(k\Delta x, n\Delta t)) \right] ds \right. \\ &\quad \left. + c \int_{n\Delta t}^{(n+1)\Delta t} (\Phi(k\Delta x, s) - \Phi(k\Delta x, n\Delta t)) dW(s) \right|^2 \\ &\leq 4a^2 \mathbb{E} \left| \int_{n\Delta t}^{(n+1)\Delta t} \left[\Phi_{xx}(k\Delta x, s) - \frac{1}{\Delta x^2} (\Phi((k+1)\Delta x, n\Delta t) \right. \right. \\ &\quad \left. \left. - 2\Phi(k\Delta x, n\Delta t) + \Phi((k-1)\Delta x, n\Delta t)) \right] ds \right|^2 \\ &\quad + 4b^2 \mathbb{E} \left| \int_{n\Delta t}^{(n+1)\Delta t} \left[\Phi(k\Delta x, s) - \frac{1}{2} (\Phi((k+1)\Delta x, n\Delta t) + \Phi(k\Delta x, n\Delta t)) \right] ds \right|^2 \\ &\quad + 4c^2 \int_{n\Delta t}^{(n+1)\Delta t} \mathbb{E} |\Phi(k\Delta x, s) - \Phi(k\Delta x, n\Delta t)|^2 ds. \end{aligned}$$

Since $\Phi(x, t)$ is a deterministic function, hence $\mathbb{E}|L(\Phi)|_k^n - L_k^n \Phi|^2 \rightarrow 0$, as $n, k \rightarrow \infty$. $\square$

Definition 4.3. *A stochastic NSFD scheme is said to be stable in the sense of mean square, if there exist some positive constants $\overline{\Delta x_0}$ and $\overline{\Delta t_0}$ and nonnegative constants K and β such that $\mathbb{E}\|u^{n+1}\|^2 \leq K e^{\beta t} \mathbb{E}\|u^0\|^2$, for all $t = (n+1)\Delta t$, $0 \leq \Delta x \leq \overline{\Delta x_0}$, and $0 \leq \Delta t \leq \overline{\Delta t_0}$.*

Theorem 4.4. *The NSFD scheme (3.2) with $t = (n + 1)\Delta t$ and $\frac{s-1}{2} \leq r \leq -s$ is stable with respect to sup-norm.*

Proof. By applying $\mathbb{E}|\cdot|^2$ in (3.2) and using the independence of the Wiener process increments, we get

$$\begin{aligned}\mathbb{E}|u_k^{n+1}|^2 &= \mathbb{E}|-ru_{k-1}^n + (1 + 2r - s)u_k^n - (r + s)u_{k+1}^n|^2 + c^2\Delta t\mathbb{E}|u_k^n|^2 \\ &\leq (|r| + |1 + 2r - s| + |r + s|^2 + c^2\Delta t) \sup_k \mathbb{E}|u_k^n|^2.\end{aligned}$$

This implies that

$$\begin{aligned}\sup_k \mathbb{E}|u_k^{n+1}|^2 &\leq (1 + c^2\Delta t) \sup_k \mathbb{E}|u_k^n|^2 \leq \dots \leq (1 + c^2\Delta t)^{n+1} \sup_k \mathbb{E}|u_k^0|^2 \\ &\leq e^{c^2\Delta t(n+1)} \sup_k \mathbb{E}|u_k^0|^2 = e^{c^2t} \sup_k \mathbb{E}|u_k^0|^2.\end{aligned}$$

So, the NSFD scheme (3.2) is stable, with $K = 1$ and $\beta = c^2$. $\square$

Definition 4.5. *A stochastic difference scheme $L_h^n u_k^n = G_k^n$ approximating the SPDE $Lu = G$ is convergent in the sense of mean square at time t , if $\mathbb{E}||u^{n+1} - v^{n+1}||^2 \rightarrow 0$, for $t = (n + 1)\Delta t$, $\Delta x \rightarrow 0$ and $\Delta t \rightarrow 0$.*

Theorem 4.6. *The NSFD scheme (3.2) for SPDE (1.1) is convergent in the sense of mean square with respect to sup-norm.*

Proof. From the stochastic Lax-Richtmyer Theorem, one can conclude that the NSFD scheme (3.2) is convergent. $\square$

5 Numerical Simulation

In this section, the theoretical results established in the preceding section are verified through one test problem. All numerical simulations were performed using MATLAB 2025 on a 64-bit Windows 11 system equipped with an Intel(R) Core(TM) i7-13800 CPU and 16 GB of RAM. The Wiener process increments were generated through the **randn** command in MATLAB.

Example 4.1. Consider the SPDE as follows

$$u_t(x, t) - u_{xx}(x, t) + u(x, t) + cu(x, t)\dot{W}(t) = 0, \quad x \in [0, 1], \quad t \in [0, 1],$$

subject to the following initial condition $u(x, 0) = \sin \pi x$ for $x \in [0, 1]$, with the boundary conditions $u(0, t) = 0$ and $u(1, t) = 0$ for $t \in [0, 1]$. It is easy to check that in the absence of the stochastic term, the exact solution is $u(x, t) = \sin \pi x e^{-(\pi^2+1)t}$.

Suppose M and N are the total number for the space and time discretizations, respectively. The NSFD scheme is conditionally stable for $\frac{s-1}{2} < r < -s$. In Figures 1–4, we compare the numerical solution produced by the proposed NSFD scheme with the exact solution for the parameter values $c = 0.5, 1, 2$, $\Delta x = 0.04, 0.05, 0.1$, and $N = 500, 2000, 2500, 6000, 10000, 15000$. For each case, 5000 sample paths were simulated.

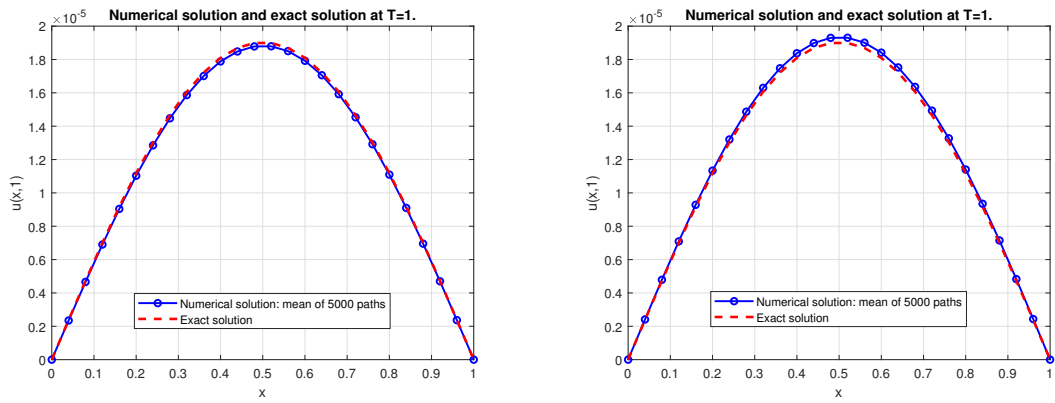


Figure 1: Comparison of the NSFD scheme with the exact solution for $c = 0.5$ and $\Delta x = 0.04$, $N = 2500$ (left), and for $c = 1$ and $\Delta x = 0.04$, $N = 2000$ (right).

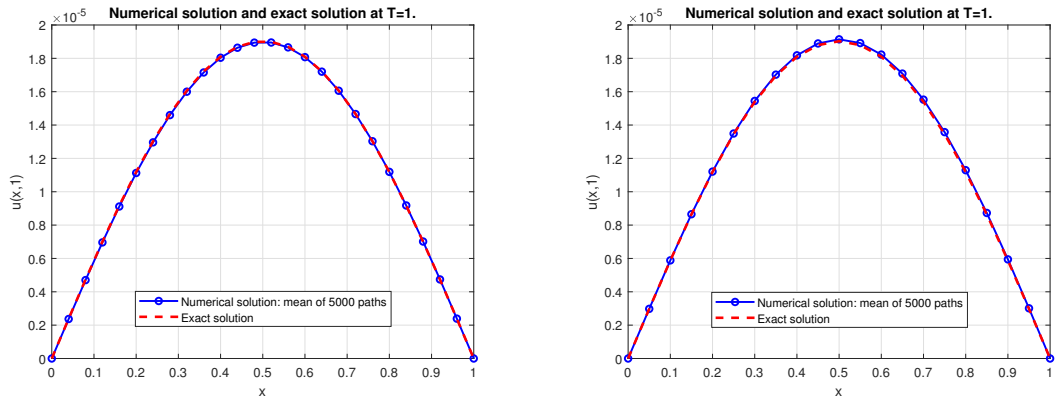


Figure 2: Comparison of the NSFD scheme with the exact solution for $c = 2$ and $\Delta x = 0.04$, $N = 5000$ (left), and for $c = 0.5$ and $\Delta x = 0.05$, $N = 5000$ (right).

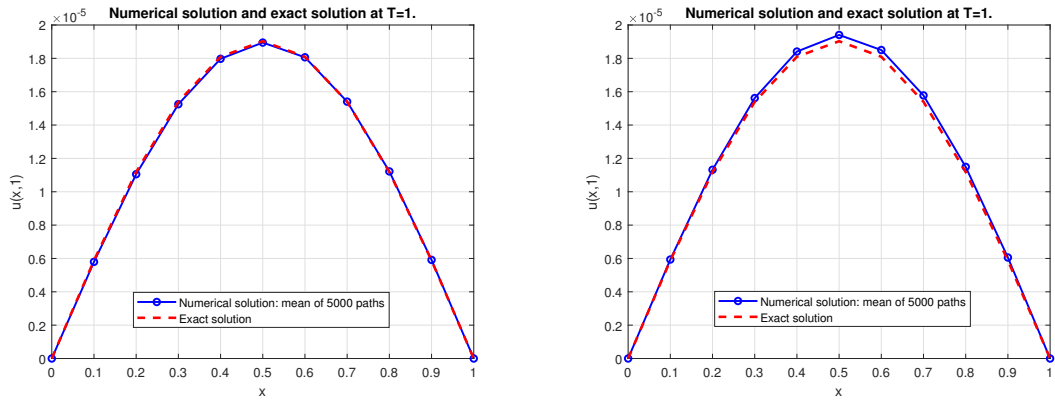


Figure 3: Comparison of the NSFD scheme with the exact solution for $c = 1$ and $\Delta x = 0.1$, $N = 500$ (left), and for $c = 2$ and $\Delta x = 0.1$, $N = 10000$ (right).

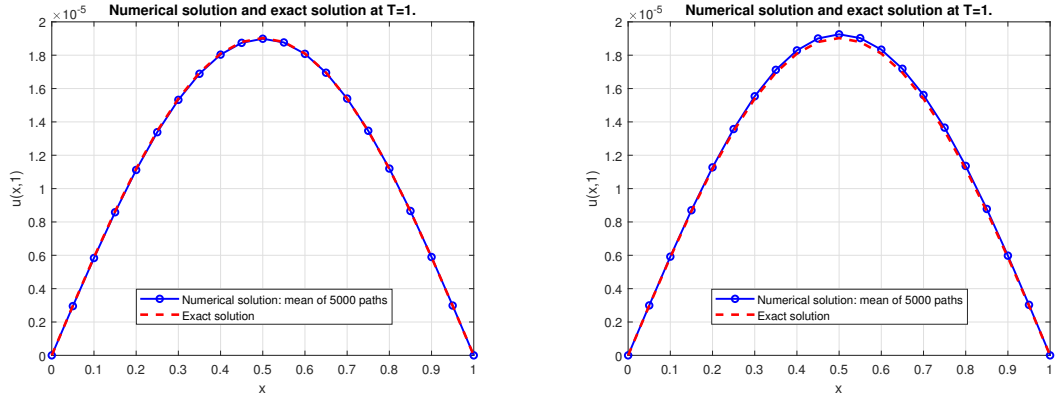


Figure 4: Comparison of the NSFD scheme with the exact solution for $c = 1$ and $\Delta x = 0.05$, $N = 6000$ (left), and for $c = 2$ and $\Delta x = 0.05$, $N = 15000$ (right).

6 Conclusions

In the present study, an NSFD scheme is proposed for the stochastic parabolic partial differential equation. The consistency, stability, and convergence of the scheme are theoretically established. The numerical study shows that the proposed NSFD scheme is a reliable and effective method for solving SPDEs.

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